

HARKIRAT SINGH

Backend Engineer & AI Systems Developer

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Professional Summary

Backend Engineer experienced in designing and shipping scalable, production-grade systems with Python, FastAPI, and relational databases. Skilled in distributed systems, RESTful API development, and clean layered architecture including caching, rate limiting, and async processing. Additional depth in Generative AI and LLM systems — multi-agent orchestration, LangChain-based RAG pipelines, and LLM-integrated backend design, with a focus on performance, concurrency safety, and maintainable code at scale.

Technical Skills

Languages	Python, SQL, Java
Backend	FastAPI, REST APIs, SQLAlchemy, Pydantic, JWT Authentication, AsyncIO
Databases	PostgreSQL, MySQL, SQLite, Redis
AI / GenAI	LangChain, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, LLM APIs, Prompt Engineering, OpenAI, Gemini
System Design	Caching, Rate Limiting, Concurrency, Scalable API Design, Clean Architecture
Developer Tools	Docker, Git, AWS (EC2, S3, RDS), Structured Logging

Internship Experience

Backend Developer Intern — AI-Powered Website SEO Audit Platform

2026

Python, FastAPI, AsyncIO, BeautifulSoup, OpenAI/Gemini APIs, REST APIs

- Engineered scalable backend services in Python and FastAPI to power an end-to-end AI-driven SEO audit platform, following a modular architecture with dedicated routers, services, crawler, prompts, reports, and database layers.
- Built an asynchronous website crawler using AsyncIO and BeautifulSoup, implementing duplicate-URL detection and robots.txt compliance to enable safe, efficient large-scale crawling.
- Developed automated Technical SEO and On-Page SEO analysis modules, extracting metadata including titles, descriptions, headings, canonical URLs, status codes, links, and page content.
- Integrated Generative AI (OpenAI and Gemini) to generate page-wise SEO scores and actionable recommendations, automating analysis that previously required manual review.
- Designed and implemented REST APIs for audit creation, audit tracking, page-level analysis, and report downloads, exposing a clean interface for frontend and third-party consumption.
- Automated generation of downloadable PDF and Excel audit reports, streamlining delivery of SEO insights to end users.
- Delivered production-ready backend code following FastAPI best practices, prioritizing modularity, readability, and scalability.

Projects

Multi-Agent Search System — LLM Orchestration Platform

2025

Python, FastAPI, LangChain, RAG, LLM APIs

- Problem: Route natural-language queries to the correct tool (search, calculator, database) reliably, with minimal hallucination and latency.
- Architected a production-grade multi-agent AI system in LangChain with a planner-executor design that dynamically routes queries to specialized tool agents via intent classification, enabling multi-step reasoning with deterministic tool sequencing.
- Implemented persistent session memory and context tracking, maintaining coherent conversation state across 10+ turn interactions.
- Added fallback and retry strategies at the agent-router level, cutting failure-induced latency spikes by ~40%; applied RAG for context injection, reducing hallucination on domain-specific queries.
- Result: Exposed the full pipeline as a FastAPI microservice with structured JSON logging, request tracing, and configurable model parameters for cost optimization.

Education

Bachelor of Engineering — Computer Engineering

2023 – 2027

Sinhgad Institute of Technology, Pune

Certifications

- Fundamentals of Machine Learning and Artificial Intelligence — AWS Training & Certification (July 2026)
- Foundations of Prompt Engineering — AWS Training & Certification (July 2026)
- Amazon Connect Routing Fundamentals — AWS Training & Certification (July 2026)